

Future of the PC

Time and Technology wait for no one. Come take a peek into our magic crystal ball as we show you what the future holds for the PC in the coming year

No more fat beige boxes, no more clutter with wires and cords, ample desk space without bulky CRTs. Is this what lies ahead?

Super powerful PCs, small and sexy form factors, thin and crisp looking TFTs, hands-free computing and wireless freedom is what we've always been promised. Now the time has come for these promises to deliver what we've always desired—a richer computing experience.

Right now we are staring straight into the end of an era in computing. Technologies and products that we are used to are almost at the end of their life-cycles and emerging technologies that are on the bleeding edge will become more common place.

By the beginning of next year we'll be ready to abandon the tried and tested interface for IDE drives. Also, most of the legacy ports that we see today at the back of our PC would be dead and buried. The PC, both internally and externally, is going to get a steroid shot in the arm that will put it firmly on the entertainment super highway. By the beginning of next year we will officially operate in multi-GHz territory and almost every device will transition to a faster and improved standard. But with this much-required facelift and makeover, the PC scene will become a little chaotic and the situation will only stabilise by mid-2003.

Inside the box

Power packed processors

In the past few months the processor manufacturing companies have only managed to get a foot in the door when it comes to GHz+ speeds. The pace has been rather slow with a lot of hiccups along the way. Starting next year, both AMD and Intel will churn out faster processors at shorter intervals, and the speeds will rise like nobody's business—imagine processors containing upwards of a billion transistors and running at speeds of over 4 GHz! Current generation CPUs are already being manufactured at an amazing 0.13 micron fabrication process and with the transition to 0.09 micron imminent, a billion transistor CPU isn't too far off—chances are you will find one in your PC next year. These GHz guzzling beasts will be powerful enough to provide as-real-as-it-gets 3D graphics, real-time capture of full-motion MPEG-2 video, and accurate voice recognition.

Bandwidth busters

New and emerging memory technologies will be quenching the thirst for continuous data. Bandwidth will double with DDR II and QBM (Quad Band Memory) taking over from the existing standards. DDR II is the next evolution of the DDR standard. It will offer double the bandwidth of standard 200 MHz DDR (PC1600) by sampling at 4-bits per cycle compared to the current 2-bits. DDR II will debut at 400 MHz (100 MHz quad-pumped) and will offer a

whopping 6.4 GBps of bandwidth under dual-channel configurations. It's expected to be widely available by the beginning of next year.

Some say that even with this insane amount of bandwidth, the next generation of CPUs could face some bottlenecks. In steps QBM or Quad Band Memory that offers twice the bandwidth of DDR. But the beauty of QBM is that it can be implemented with any of today's standard DDR modules. With a QBM memory controller, a current generation DDR266 memory module (PC2100) will deliver 4.2 GBps, which is double its theoretical bandwidth. Compared to DDR II, QBM can scale faster, reaching PC5400 levels with existing DDR333 modules. Whatever the memory technology, processors won't have a reason to complain next year.

Ga-Ga graphics

With bandwidth-consuming texture-rich games becoming more and more common, the current graphics platform will have to go through a serious upgrade to keep up. Also, every graphics card that will be released next year from nVidia and ATI will have integrated TnL (Transform and Lighting) along with programmable shaders support. The technology of shaders is being used in almost all new games that are being developed, which can cause the video memory bandwidth to become a severe bottleneck. This would be alleviated to a large degree with the introduction of AGP 8x.

The official name for AGP 8x is now